$$\mathbb{Z}/2 \;\to\; S^\infty \;\to\; \mathbb{R}P^\infty$$

$$\mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \leftarrow \cdots$$